

AUTOMATED SOFTWARE ENGINEERING

MEI 2026/27

PL Classes #4
Docker

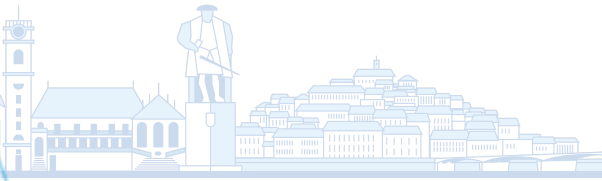
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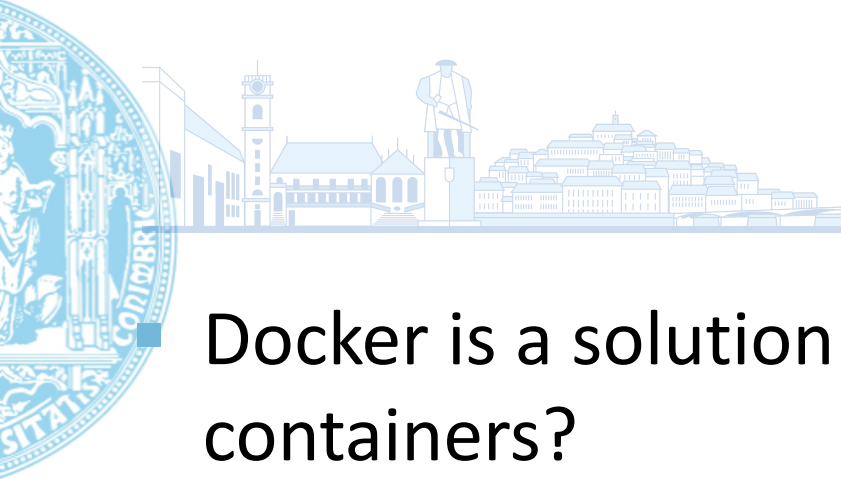


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SOFTWARE ENGINEERING

DOCKER





■ Docker is a solution for containers, but what are containers?

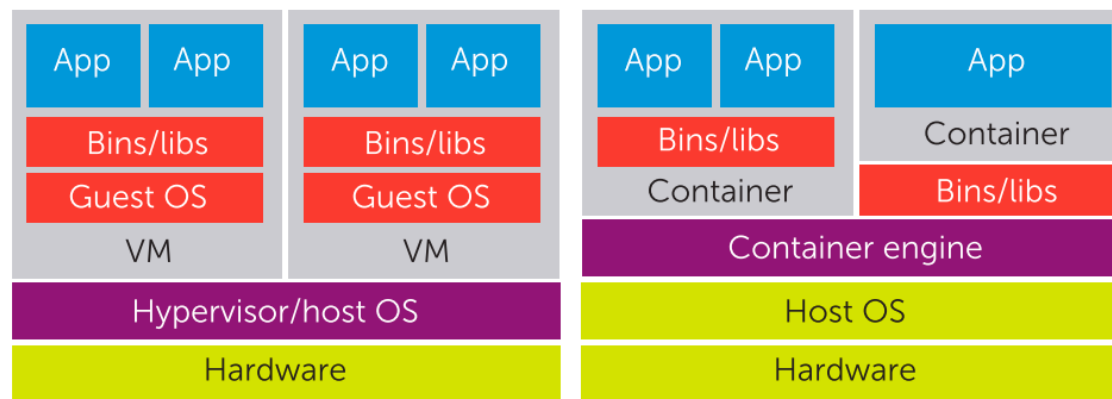


FIGURE 1. Virtualization architecture. The two possible scenarios, a traditional hypervisor architecture on the left and a container-based architecture on the right, differ in their management of guest operating system components.

C. Pahl, "Containerization and the PaaS Cloud," in IEEE Cloud Computing, vol. 2, no. 3, pp. 24-31, May-June 2015, doi: 10.1109/MCC.2015.51.



DOCKER

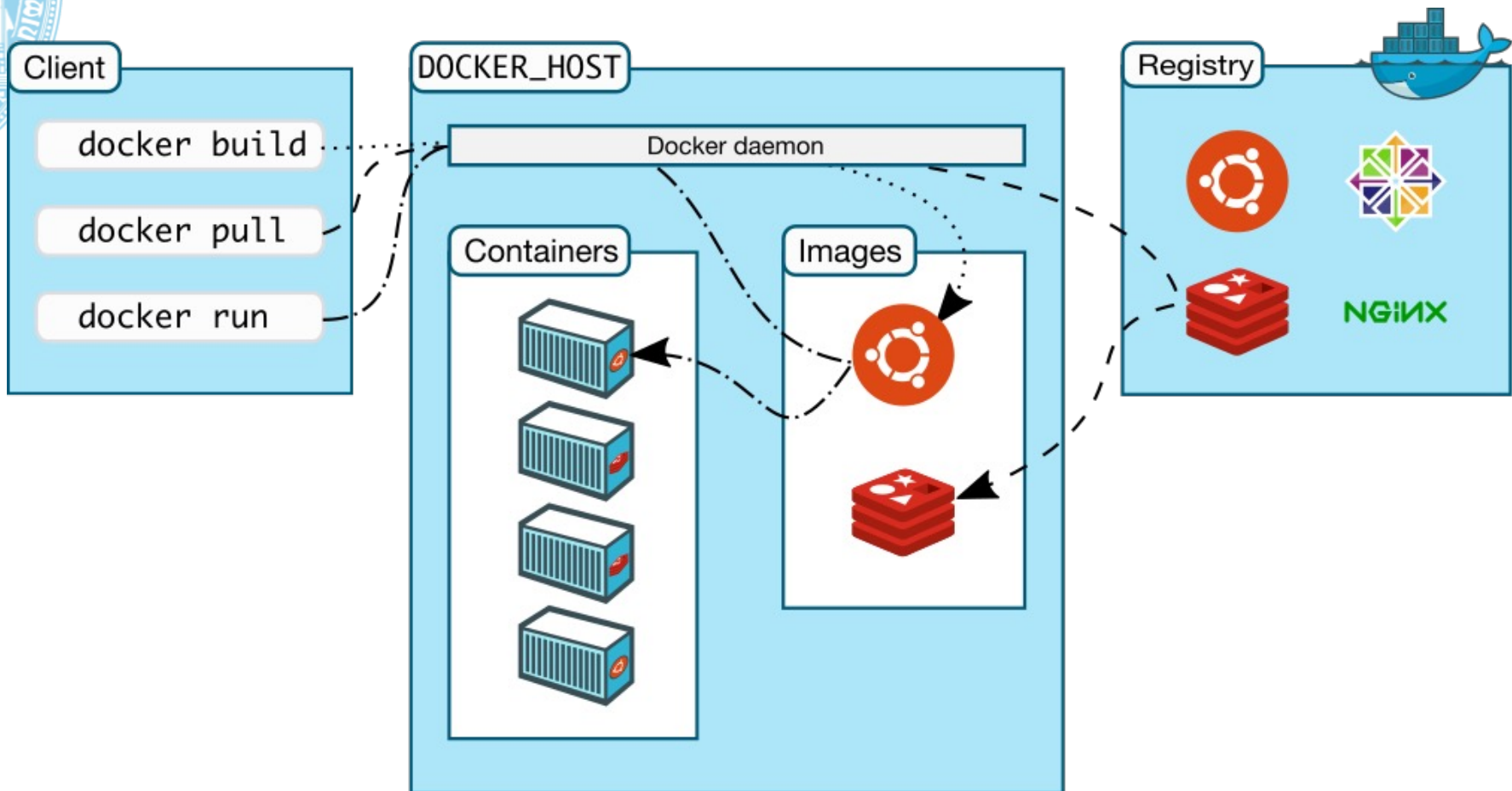
- Container solution created in 2013
- Established the OCI (Open Container Initiative) with other container market leaders
- Widely used to deploy microservice applications
- Allows generating “images” of containers and also execute them
- Provides a solution for container orchestration

<https://www.docker.com>





DOCKER ARCHITECTURE



Source: <https://docs.docker.com/get-started/overview/>



DOCKERFILE AND IMAGE CREATION

- Used to define how an image should be created

```
# syntax=docker/dockerfile:1
FROM node:12-alpine
RUN apk add --no-cache python2 g++ make
WORKDIR /app
COPY . .
RUN yarn install --production
CMD ["node", "src/index.js"]
EXPOSE 3000
```

- To create the image

```
$ docker build -t getting-started .
```

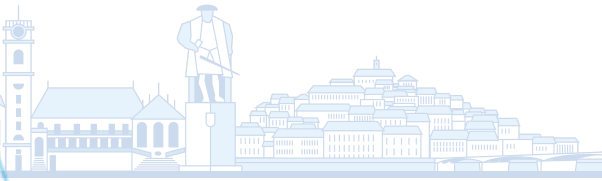
https://docs.docker.com/get-started/02_our_app/



MAIN DOCKER COMMANDS

Docker CLI:

- `docker version`
- `docker images`
- `docker image list`
- `docker search`
- `docker image pull`
- `docker run [IMAGE_NAME]`
- `docker run -p [PORT]`
`[IMAGE_NAME]`
- `docker ps`
- `docker ps -a`
- `docker stats [CONTAINER_ID]`
- `docker inspect`
- `docker rmi [IMAGE_NAME]`
- `docker exec`
- `docker start`
- `docker stop`



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KUBERNETES



KUBERNETES

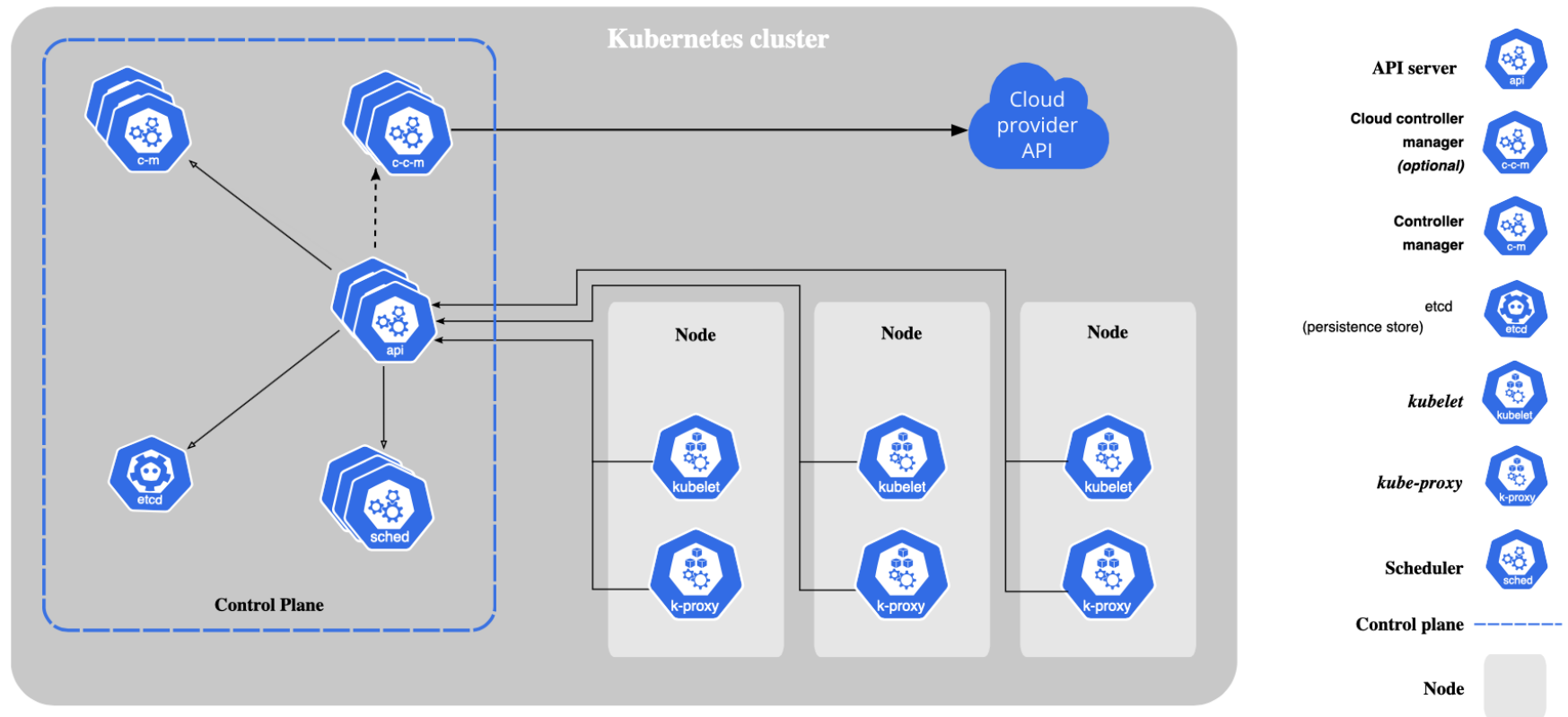
- Most used solution for container orchestration
 - Mostly Docker containers
- Created at Google in 2014, but maintained by the community (Cloud Native Computing Foundation)
- Each container is wrapped in a “pod”, which is the smallest control unit of Kubernetes
- The cluster is composed of a control-plane and nodes

Source: <https://kubernetes.io>





KUBERNETES ARCHITECTURE



<https://kubernetes.io/docs/concepts/overview/components/>

QUESTIONS?



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